Shoumik Saha

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/ShoumikSaha | Google Scholar

RESEARCH INTEREST

- **Reliable AI** - Security & Reliability of GenAI, LLM Alignment & Hallucination
- **Computer Security** - Adversarial Attacks & Defenses, Malware

WORK EXPERIENCE

Applied Scientist II (L5) — **Sep 2026 – Present**
Amazon AWS — *Full Time*
Managers - Varun Kumar, Aritra Sengupta

Graduate Research Assistant — **Aug 2023 – Aug 2026**
Department of Computer Science — *Full Time*
University of Maryland
Supervisor - Dr. Soheil Feizi

Applied Scientist Intern — **May 2025 – Aug 2025**
Amazon AWS — *Full Time*
Managers - Varun Kumar, Zijian Wang

Applied Scientist Intern — **Jun 2024 – Aug 2024**
Amazon AWS — *Full Time*
Manager - Zijian Wang

Graduate Research Assistant — **Aug 2022 – Jul 2023**
Maryland Cybersecurity Center — *Full Time*
Supervisor - Dr. Tudor Dumitras

Lecturer — **Jul 2021 – Jul 2022**
Department of Computer Science and Engineering — *Full Time*
United International University

Research Assistant — **Mar 2021 – Jul 2022**
Data Science and Engineering Research Lab — *Part Time*
Bangladesh University of Engineering and Technology (BUET)

EDUCATION

University of Maryland - College Park (UMD) — **Aug 2022 – Aug 2026**
Ph.D. in Computer Science — *CGPA – 3.814/4.0*
Advisor – Dr. Soheil Feizi

University of Maryland - College Park (UMD) — **Aug 2022 – Aug 2024**
M.S. in Computer Science
Advisor – Dr. Soheil Feizi

Bangladesh University of Engineering and Technology (BUET) — **Mar 2016 – Feb 2021**
B.Sc. in Computer Science and Engineering — *Advanced GPA – 3.79/4.0*
Thesis supervisor – Dr. Atif Hasan Rahman

PUBLICATIONS

1. **Under the Hood of SKILL.md: Semantic Supply-chain Attacks on AI Agent Skill Registry**
   [**Shoumik Saha**, K Faghih, S Feizi]
   Under Review (arXiv link)
   Media Coverage: The Register

2. **Breaking the Code: Security Assessment of AI Code Agents Through Systematic Jailbreaking Attacks**
[**Shoumik Saha**, J Chen*, S Mayers*, SK Gouda, Z Wang, V Kumar]
TACL 2026(*Transactions of the Association for Computational Linguistics*) (Paper Link)

3. **Almost AI, Almost Human: The Challenge of Detecting AI-Polished Writing**
[**Shoumik Saha**, S Feizi]
ACL 2025 (*Association for Computational Linguistics*) (Paper Link)
Media Coverage: The New York Times, Plagiarism Today, The Science Archive, The Cheat Sheet

4. **Adversarial Paraphrasing: A Universal Attack for Humanizing AI-Generated Text**
[Y Cheng, VS Sadasivan, **Shoumik Saha***, M Saberi*, S Feizi]
NeurIPS 2025 (*Conference on Neural Information Processing Systems*) (Paper Link)

5. **ProcVQA: Benchmarking the Effects of Structural Properties in Mined Process Visualizations on Vision–Language Model Performance**
[KT Zinat*, SM Abrar*, **Shoumik Saha**, S Duppala, SN Sakhamuri, Z Liu]
EMNLP 2025 (*Empirical Methods in Natural Language Processing*) (Paper Link)

6. **LLM-Check: Investigating Detection of Hallucinations in Large Language Models**
[G Sriramanan, S Bharti, VS Sadasivan, **Shoumik Saha**, P Kattakinda, S Feizi]
NeurIPS 2024 (*Conference on Neural Information Processing Systems*) (Paper Link)

7. **Fast Adversarial Attacks on Language Models In One GPU Minute**
[VS Sadasivan, **Shoumik Saha***, G Sriramanan*, P Kattakinda, A Chegini, S Feizi]
ICML 2024 (*International Conference on Machine Learning*) (Paper Link)
Media Coverage: The Register

8. **DRSM: De-Randomized Smoothing on Malware Classifier Providing Certified Robustness**
[**Shoumik Saha**, W Wang, Y Kaya, S Feizi, T Dumitras]
ICLR 2024 (*International Conference on Learning Representations*) (Paper Link)

9. **MAlign: Explainable Static Raw-byte Based Malware Family Classification using Sequence Alignment**
[**Shoumik Saha**, S Afroz, A Rahman]
Computers & Security Journal (Paper Link)

10. **ML-Based Behavioral Malware Detection Is Far From a Solved Problem**
[Y Kaya, Y Chen, Marcus Botacin, **Shoumik Saha**, F Pierazzi, L Cavallaro, D Wagner, T Dumitras]
SaTML 2025 (*Secure & Trustworthy Machine Learning*) (Paper Link)

SCHOLARSHIPS & AWARDS

- Awarded *'Dean's Fellowship'* from the University of Maryland
- Awarded *'Innovation Fund'* for research from the ICT division of Bangladesh government
- Achieved *'Merit Stipends'* from BUET in five out of seven terms
- Awarded *'Dean's Award'* in Junior year for extra-ordinary result
- Achieved *'Talent-Pool Scholarship'* in High School

TEACHING

Teaching Assistant
University of Maryland *Fall 2023*
Data 200: Knowledge in Society: Science, Data and Ethics

Lecturer
United International University *Fall 2021 - Summer 2022*
Discrete Mathematics, Data Structure and Algorithm, Operating Systems

SERVICES

- **Reviewer**, top-tier ML conferences such as ICML 2024, ICLR 2025, NEURIPS 2025, etc.
- **External Reviewer**, top-tier Computer Security conference, including USENIX 2023
- **Lab Instructor**, United International University
- **General Secretary**, BUET Photographic Society